Math 133 Written Homework 6

Differentiation Rules and Graphs of Derivatives

Due Thursday, October 22, 2026, at 11:59 p.m.

Show your work, including the differentiation rules you use and reasons for your conclusions. You may use differentiation rules unless the limit definition is requested. Include units when interpreting business quantities and rates of change. Label axes and mark important points on your graphs.

List collaborators, time spent, and any use of generative AI with your submission. Your submitted work must be your own. Late work will not be accepted.

1. Let $p(q)$ be the price, in dollars per item, at which a company can sell exactly q items. Suppose p is differentiable near 85, with

$$p(85) = 24, \qquad p'(85) = -0.12.$$

   Write the revenue function $R(q)$ in terms of $p(q)$. Compute $R(85)$ and $R'(85)$. (Hint: Use the product rule to compute the derivative.) Use the information given to estimate $R(90)$ as accurately as you can.

2. Differentiate each function.

   (a) $f(x) = 4x^5 - 3x^3 + 6x - 8.$

   (b) $g(x) = (x^2 + 1)(3x - 2).$

   (c) $h(x) = 6\sqrt{x} - \dfrac{4}{x} + \dfrac{3}{x^2}.$

3. Differentiate each function, and state where its derivative is defined.

   (a) $u(x) = \dfrac{2x^2 - 3x + 1}{x + 2}.$

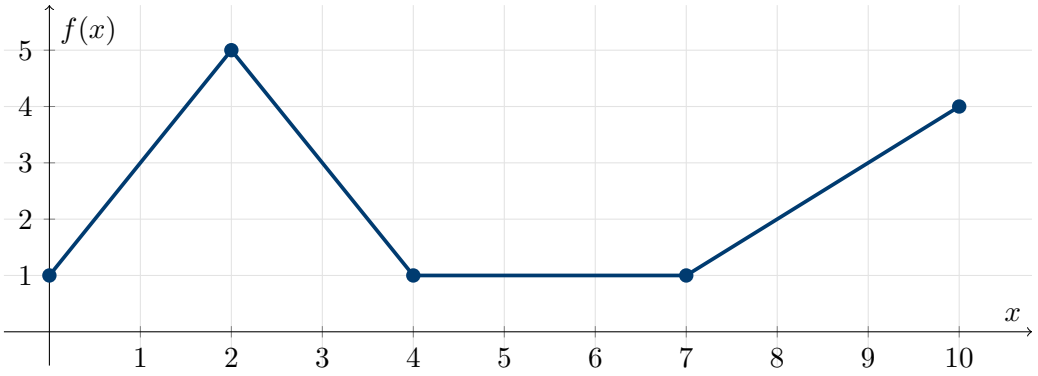
   (b) $v(x) = \dfrac{x^3 + 4}{x^2 + 1}.$

   (c) $w(x) = \dfrac{\sqrt{x}}{x + 4}.$

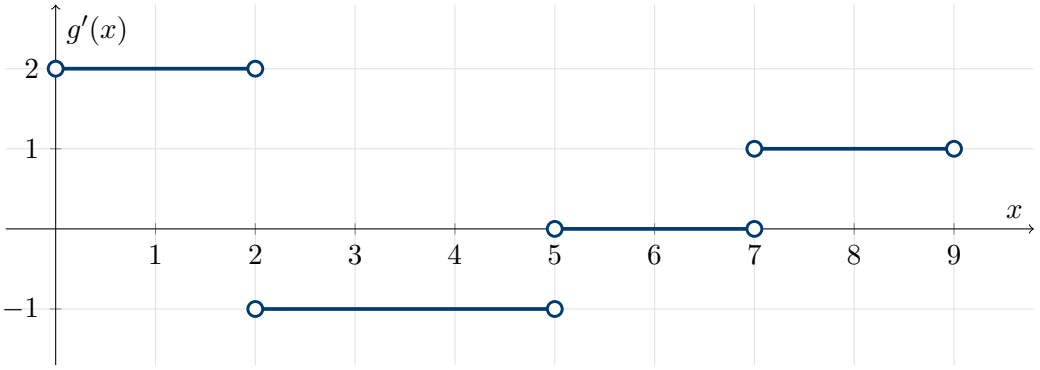
4. A company's total cost is $C(q)$ dollars for q items. Its average cost per item is $A(q) = C(q)/q$, for $q > 0$. Suppose C is differentiable near 80, with $C(80) = 2400$ and $C'(80) = 22$. Find $A(80)$ and $A'(80)$. Is average cost increasing or decreasing at $q = 80$? Explain in the business context.

5. Let $a(x) = |x|$. Compute $a'(x)$ wherever it exists, and state the domains of a and a'. Is $a'(0)$ defined? Explain why or why not using the limit definition of the derivative.

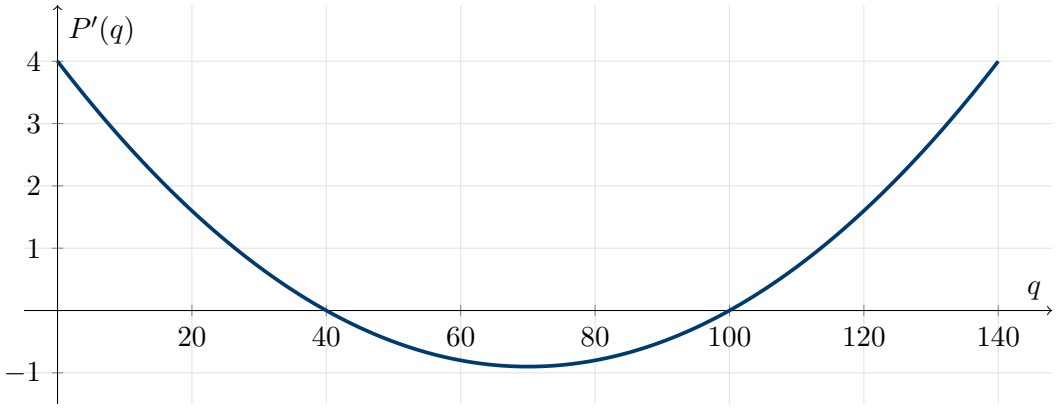
6. The graph of a continuous piecewise linear function f is shown below. Sketch the graph of f' on $0 < x < 10$, and state every point in this interval where f' does not exist.



7. The graph below gives g' wherever it exists on $(0, 9)$. Sketch two functions g on $[0, 9]$ with this derivative and $g(0) = 1$: one continuous and one discontinuous.



8. A company's profit $P(q)$ is a smooth function of the number q of items produced and sold, for $0 \le q \le 140$. Treat q as continuous. The graph shows the marginal profit $P'(q)$, measured in dollars per item.



(a) On what intervals is profit increasing or decreasing?

(b) At which quantities in $(0, 140)$ does profit have a local maximum or a local minimum? Justify your answers using the graph.

(c) If the company currently sells 60 items, should it increase or decrease sales slightly to improve profit? What if it currently sells 120 items?

(d) At $q = 80$, the graph of P' is rising. Is profit increasing there? Explain. Can you tell from this graph whether the company's profit at $q = 40$ is positive?